SOFTWARE REQUIREMENT SPECIFICATION

SRS Document

AI-Smart Campus System

Intelligent Academic Assistance Ecosystem

Team ID	CSE4204-8A-T07
Course	CSE 4204 — Mobile Computing Lab
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1. Introduction

1.1 Project Title

Project Name	AI-Smart Campus System

1.2 Project Overview

The AI-Smart Campus System is an intelligent web-based platform tailored for Northern University of Business and Technology, Khulna (NUBT Khulna). It bridges the operational gaps found in traditional, passive campus portals by introducing:

- Automated performance metrics and CGPA tracking.
- Algorithmic elective track selection via AI Course Advisor.
- A 24/7 conversational campus assistant powered by the Gemini API.
- Proactive at-risk student detection using rule-based backend analysis.

The system uses a React.js and Tailwind CSS presentation tier coupled with an enterprise-grade Laravel backend running over a local MySQL database.

The proposed system integrates Artificial Intelligence to transform a conventional campus platform into an intelligent academic assistance ecosystem. It connects directly with institutional academic data, student performance records, and official notices to provide accurate, campus-specific guidance.

1.3 Problem Statement

Modern university students often face significant difficulties in managing fragmented academic information, tracking personal performance indicators, and staying up-to-date with institutional notices. Faculty members struggle to efficiently monitor student engagement. Traditional Learning Management Systems (LMS) function primarily as passive information repositories, lacking intelligent automation.

In many educational institutions, student performance is monitored manually through attendance records, quizzes, assignments, and examination results. Although large amounts of academic data are collected, very little of it is used for predictive analysis or personalized guidance.

As a result, students often fail to recognize academic difficulties early, while faculty members face challenges in identifying learners who require additional support. Existing systems provide limited assistance in predicting academic risks or recommending suitable academic pathways.

Therefore, an intelligent system is needed to continuously analyze academic performance, detect potential risks, provide personalized recommendations, and support timely academic intervention.

1.4 Objectives

- To build a smart, all-in-one student productivity portal connected to campus workflows.
- To generate personalized study paths based on historical grading and performance trends.
- To maximize student success through predictive grade warnings.
- To integrate a 24/7 AI Academic Assistant (Chatbot) for institutional information access.
- Develop a centralized platform for academic performance monitoring.
- Predict academically at-risk students using AI techniques.
- Generate real-time alerts for attendance and academic issues.
- Support faculty members in student evaluation and monitoring.
- Improve overall academic performance and decision-making.

1.5 Scope of the Project

The system covers role-based access for students and faculty. It features an AI-powered course advisor, student at-risk detection analytics, and an NLP-driven chatbot for automated campus assistance.

The system focuses on academic performance analysis, risk prediction, course recommendation, and AI-assisted academic support.

Students will be able to monitor their performance, receive recommendations, obtain risk alerts, and communicate with the AI assistant. Faculty members will track student progress and identify at-risk learners, while administrators will manage users and oversee platform operations.

The current version does not include online examinations, automated grading systems, or external LMS integration.

2. Functional Requirements

The following functional requirements define the core capabilities of the AI-Smart Campus System:

Req. ID	Requirement Name	Description
FR-01	Authentication	Students, Faculty, and Admin can securely log in using JWT authentication.
FR-02	Student Dashboard	Students can view attendance, quiz, assignment, and mid-term performance through visual charts.
FR-03	Attendance Monitoring	The system tracks and displays attendance records.
FR-04	Academic Performance Analysis	The system analyzes quiz, assignment, and examination results.
FR-05	AI Course Recommendation	Students receive personalized core and elective course suggestions.
FR-06	Risk Prediction	The system identifies students who may be academically at risk.
FR-07	Risk Alert Notification	Students receive warnings for low attendance or poor performance.
FR-08	Faculty Monitoring	Faculty members can monitor student performance and engagement.
FR-09	Academic Evaluation	Faculty members can evaluate students using performance reports.
FR-10	AI Chatbot Support	Users can interact with the Gemini-powered chatbot.
FR-11	Academic Information Service	The chatbot provides notices, routines, exam schedules, and study guidance.
FR-12	User Management	Admin can approve, update, and manage student and faculty accounts.
FR-13	System Monitoring	Admin can monitor platform activities and reports.

3. Non-Functional Requirements

Non-functional requirements define the quality attributes and system constraints:

Req. ID	Requirement Name	Specification
NFR-01	**Performance**	System pages shall load within 3 seconds under normal conditions.
NFR-02	**Security**	JWT authentication must be implemented.
NFR-03	**Data Protection**	User credentials must be securely encrypted.
NFR-04	**Responsiveness**	The system must support desktop, tablet, and mobile devices.
NFR-05	**Reliability**	The system should maintain high availability and stability.
NFR-06	**Scalability**	The architecture should support future expansion.
NFR-07	**Usability**	The interface should be simple and user-friendly.
NFR-08	**Maintainability**	The codebase should follow modular development practices.

4. User Roles

The system supports three distinct user roles, each with specific permissions and access levels:

Role	Responsibilities
Admin	• Manage overall system operations. • Approve and manage student accounts. • Approve and manage faculty accounts. • Monitor platform activities. • Generate administrative reports. • Maintain security and access control.
Faculty	• Login securely. • Monitor attendance and academic performance. • Analyze quiz, assignment, and examination results. • Identify at-risk students. • Conduct academic evaluations. • Review AI-generated performance reports.
Student	• Login securely. • View academic performance dashboard. • Monitor attendance and grades. • Receive AI-generated course recommendations. • Receive academic risk alerts. • Communicate with AI chatbot. • Access notices, schedules, and study plans.

5. Use Cases

The following use cases describe the primary interactions between users and the AI-Smart Campus System:

Use Case 1: User Login
Actor: Student / Faculty / Admin
Action: User enters credentials and submits login request.
System Response: System validates credentials and grants access to the appropriate dashboard.

Use Case 2: Identify At-Risk Students
Actor: Student
Action: Student accesses the dashboard.
System Response: System displays attendance, quiz, assignment, and examination statistics using visual charts.

Use Case 3: Receive Course Recommendations
Actor: Student
Action: The student requests course suggestions.
System Response: AI analyzes academic records and recommends suitable courses.

Use Case 4: Receive Risk Alert
Actor: Student
Action: Performance or attendance falls below the threshold.
System Response: System generates warning notifications and improvement suggestions.

Use Case 5: Use AI Chatbot
Actor: Student
Action: User submits an academic query.
System Response: Gemini-powered chatbot provides relevant responses and guidance.

Use Case 6: Monitor Student Performance
Actor: Faculty
Action: Faculty accesses student analytics.
System Response: System displays performance reports, trends, and risk indicators.

Use Case 7: Manage Users
Actor: Admin
Action: Admin manages student and faculty accounts.
System Response: System updates user records and permissions accordingly.